Discrete Mathematics – Final Exam

2022 – Moed A

Instructions for students:

1. In the exam there are **12** short questions and **2** open questions. You must answer **all** of them (No choice).
2. Write your solutions in the attached papers only.
3. When asked to write explicitly a "set/function/relation", you must specify all the elements of the set/function/relation.
4. Use of any external aiding material is **forbidden**. For your convenience, a formula sheet is attached.
5. The duration of the exam is **three hours.**

Good luck!

Part A – Short Questions

- Write your answers clearly in the marked boxes.
- No need to explain how you got to the final answer, **unless it is explicitly asked**.

In counting questions, you can use any of the convenient notations which we used in class, **except** for the symbol $\sum$ and three dots, '…'.

Question 1 (6 points) – Simplify the definition.

Provide an example that disproves the following claim (a counter example):

For any 2 sets S, T, if $S \cap P(S) \neq \emptyset$ and $S \cap P(T) \neq \emptyset$, then $S \cap T \neq \emptyset$.

No need to explain, just write down the sets.

$$S = \{\emptyset\}, T = \{1\}$$

Question 2 (6 Points)

The connective $*$ is defined by the following truth table:

p	q	$q * p$
T	T	F
T	F	F
F	T	T
F	F	F

Write a short statement that is equivalent to $q * p$, including only the connectives $\neg, \wedge$.

$$q * p \equiv \neg(q \rightarrow p) \equiv q \wedge \neg p$$

Question 3 (6 points).

Let A be the set of all total functions from $\mathbb{N}$ to $\mathbb{N}$.

Define $h: A \times \mathbb{N} \times \mathbb{N} \to \mathbb{N} \times \mathbb{N}$ by $h(f, n, m) = (f(n), f(m))$ (note that f is a function from $\mathbb{N}$ to $\mathbb{N}$). Find a <u>right inverse</u> of f (no need to prove).

Let $g: \mathbb{N} \times \mathbb{N} \to A \times \mathbb{N} \times \mathbb{N}$ be $g(n, m) = (i_{\mathbb{N}}, n, m)$

Question 4 (6 points).

What is the coefficient of x^8 in the expansion of the expression $(1 + x + x^3)^{10}$?

$$\binom{10}{2,8} + \binom{10}{1,5,4} + \binom{10}{2,2,6}$$

Question 5 (6 points)

Let $n \geq 1$ be a natural. How many ways are there to distribute n identical white balls and $2n$ identical black balls in n cells such that in any cell, the number of black balls is greater or equal the number of white balls?

$$\binom{2n-1}{n}^2$$

Question 6 (6 points)

How many ways are there to build a 6 characters long password, using the characters $1,2,3,4,5,6,a,b,c,d,e,f$ such that the all the characters are different and the password includes at least 2 letters and at least 2 digits?

$$\binom{6}{2}\binom{6}{4}6! + \binom{6}{3}\binom{6}{3}6! + \binom{6}{4}\binom{6}{2}6!$$

Question 7 (6 points)

How many natural solutions to the equation $x_1 + x_2 + x_3 + x_4 + x_5 = 10$ are there, such that no x_i equals 3?

$$\binom{14}{4} - \binom{5}{1}\binom{10}{3} + \binom{5}{2}\binom{6}{2} - \binom{5}{3}\binom{2}{1}$$

Question 8 (6 Points)

Provide a set A and a relation R over A that is not symmetric, not anti-symmetric, but is a (total) left invertible function from A to A.

$$A = \{1,2,3,4,5\}$$
$$R = \{(1,2),(2,1),(3,4),(4,5),(5,3)\}$$

Question 9 (6 Points).

Define a equivalence relation R over $P(\mathbb{N})$ by

$$ARB \quad \Leftrightarrow \quad (A \setminus \{1,2\} \neq \emptyset \ \ \wedge \ \ B \setminus \{1,2\} \neq \emptyset) \vee (A = B).$$

Is $\{3\}R\{4,5\}$? Is $\{3\}R\{1,2\}$?

Describe $[\mathbb{N}_{odd}]_R$ and $[\emptyset]_R$.

Yes.

No.

$$[\mathbb{N}_{odd}]_R = P(\mathbb{N}) \setminus P(\{1,2\})$$

$$[\emptyset]_R = \emptyset$$

Question 10 (6 points)

Let $A = \{x \in \mathbb{Z} \mid |x| \leq 2\}$.

Define R to be the following strong partial order relation:

$$(a,b)R(c,d) \quad \Leftrightarrow \quad |a| + |b| < |c| + |d|$$

Is R total? Write the minimal elements, the maximal elements, the minimum and the maximum (if exist).

No.

Minimum + only minimal: $(0,0)$

Maximals: $(\pm 2, \pm 2)$

There is no maximum.

Question 11 (6 points)

Ziv, the tutor, was trying to write the following question for the test:

“17 days out of the 20 last days were rainy. Use the pigeonhole principle to prove that there were at least ??? straight days.”

What is the maximal value of ??? such that the students can still prove the claim?

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Question 12 (6 points)

Student A and student B want the study CS in IDC. Each of them has two options: the Hebrew track and the international track. In addition, both tracks rank the students. Provide an explicit example of preference lists for the students (that rank the tracks) and for the tracks (that rank the students), such that the matching will be stable, no matter what side is the "suitor". In addition, write the stable matching.

Student A: Hebrew, international	Hebrew: A, B
Student B: international, Hebrew	international: B, A
The matching: A – Hebrew, B - international	

Part B – Open Questions

- Write your answers clearly in the marked boxes. You must prove your answers.
- **Write clear and readable answers! Your grade will also depend on the clarity of your proof and not only the correctness of your answer!**

Question 13 (14 points)

a. Disprove: If $A \nsubseteq B \cap C$ then $(A \setminus B) \cap (A \setminus C) \neq \emptyset$.

b. Prove: If $A \nsubseteq B \cap C$ then $(A \setminus B) \cup (A \setminus C) \neq \emptyset$.

a. Let $A = \{1,2\},\ B = \{1\}, C = \{2\}$.
So $A = \{1,2\} \nsubseteq \emptyset = \{1\} \cap \{2\} = B \cap C$,
But $(A \setminus B) \cap (A \setminus C) = (\{1,2\} \setminus \{1\}) \cap (\{1,2\} \setminus \{2\}) = \{2\} \cap \{1\} = \emptyset$.

b. Prove by contra-positive:
Assume $(A \setminus B) \cup (A \setminus C) = \emptyset$ and prove $A \subseteq B \cap C$.
Let $x \in A$. Assume by contradiction that $x \notin B$.
By definition of difference, we get $x \in A \setminus B$, and by definition of union, we get $x \in (A \setminus B) \cup (A \setminus C) = \emptyset$ a contradiction (the empty set has no elements).
Hence $x \in B$. One can easily adapt the proof to get $x \in C$.
By definition of intersection, we get $x \in B \cap C$.
Hence $A \subseteq B \cap C$.

Question 14 (14 points)

Let R be an equivalence relation over a set A. Let $R^C = \{(a,b) \in A \times A \mid (a,b) \notin R\}$ be the complement of R. Assume R^C is transitive. Prove $\left|{}^{A}/_{R}\right| = 1$.

Let $a, b \in A$. Assume by contradiction that $\neg aRb$.

If bRa then aRb by symmetry (recall that R is an equivalence relation and therefore symmetric), so we must have $\neg bRa$.

Hence $aR^C b$ and $bR^C a$, which implies $aR^C a$ due to the transitivity of R^C.

Hence $\neg aRa$, which contradicts the fact that R is reflexive as an equivalence relation.

Hence aRb for any $a, b \in A$,

so $A \times A$ is the only equivalence class and $|A/R| = |\{A \times A\}| = 1$.

An extra page for the open questions:

Discrete Mathematics – Formula sheet for the exam

Set Theory:

$$A\Delta B = (A\backslash B) \cup (B\backslash A)$$
$$A \times B = \{(a,b)|\ a \in A\ and\ b \in B\}$$
$$P(A) = \{S|S \subseteq A\}$$

$$|A \times B| = |A| \cdot |B|$$
$$|P(A)| = 2^{|A|}$$

De-Morgan:

$$\overline{B \cup C} = \bar{B} \cap \bar{C} \qquad A\backslash(B \cup C) = (A\backslash B) \cap (A\backslash C)$$
$$\overline{B \cap C} = \bar{B} \cup \bar{C} \qquad A\backslash(B \cap C) = (A\backslash B) \cup (A\backslash C)$$

Distributive:

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Functions/Relations:

$$i_A = \{(a,a)|a \in A\}$$

Logic:

For all formulas α, β, γ :

De-Morgan:

$$\neg(\alpha \wedge \beta) \equiv \neg\alpha \vee \neg\beta$$
$$\neg(\alpha \vee \beta) \equiv \neg\alpha \wedge \neg\beta$$

Distributive:

$$\alpha \wedge (\beta \vee \gamma) \equiv (\alpha \wedge \beta) \vee (\alpha \wedge \gamma)$$
$$\alpha \vee (\beta \wedge \gamma) \equiv (\alpha \vee \beta) \wedge (\alpha \vee \gamma)$$

Negation rules:

$$\neg\forall x(\alpha) \equiv \exists x(\neg\alpha)$$
$$\neg\exists x(\alpha) \equiv \forall x(\neg\alpha)$$

Combinatorics:

Choosing k items out of n different items:

	With repetition	Without repetition
With order	n^k	$\frac{n!}{(n-k)!}$
Without order	$\binom{k+n-1}{n-1}$	$\binom{n}{k} = \frac{n!}{(n-k)! \cdot k!}$

Binomial Formula

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} \cdot x^k y^{n-k}$$

Multinomial Coefficient:

$$\binom{n}{i_1, i_2, i_3, \dots, i_k} = \frac{n!}{i_1! \cdot i_2! \cdot i_3! \cdot \dots \cdot i_k!} \quad if \;\; i_1 + i_2 + i_3 + \cdots + i_k = n$$

Multinomial Formula:

$$(x_1 + x_2 + x_3 + \cdots + x_k)^n = \sum_{i_1+i_2+i_3+\cdots+i_k=n} \binom{n}{i_1, i_2, i_3, \dots, i_k} \cdot x_1^{i_1} \cdot x_2^{i_2} \cdot x_3^{i_3} \cdot \dots \cdot x_k^{i_k}$$

Pascal's Triangle:

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

$$\binom{n}{k} = \binom{n}{n-k}$$

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$$

Inclusion-exclusion principle:

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{I \subseteq \{1,2,\dots,n\},\;\; I \neq \emptyset} (-1)^{|I|-1} \cdot \left| \bigcap_{\mathrm{i} \in \mathrm{I}} \mathrm{A_i} \right|$$

Generalized Pigeonhole Principle:

If we put m pigeons in n pigeonholes, then there is at least one pigeonhole in which there will be at least $\left\lceil \frac{m}{n} \right\rceil$ pigeons.